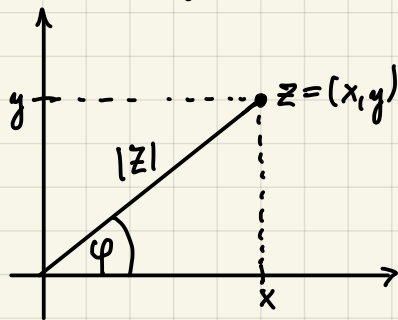


Teoria

$$Z = x + jy = |Z| \cdot (\cos \varphi + j \sin \varphi)$$



$$Z = x + jy$$

$$\bar{Z} = x - jy \rightarrow \text{sprzężenie } Z$$

$$|Z| = \sqrt{x^2 + y^2} \in \mathbb{R}$$

φ - argument Z

$$\varphi \begin{cases} \cos \varphi = \frac{x}{|Z|} \\ \sin \varphi = \frac{y}{|Z|} \end{cases} \Rightarrow \exists! \varphi \in [0; 2\pi)$$

$$j^2 = -1 \quad \blacksquare$$

Przy

$$\frac{3-j}{1+j} \cdot \frac{1-j}{1-j} = \frac{(3-j)(1-j)}{1^2 - j^2} = \frac{3 - 4j + j^2}{1^2 - j^2} = \frac{2-4j}{2} = 1-2j \quad \begin{matrix} \text{Re}z = 1 \\ \text{Im}z = -2 \end{matrix}$$

Zad. 1

$$a) (3+2j)(5-2j) = 15 + 4j - 4j^2 = 19 + 4j \quad \begin{matrix} \text{Re}z = 19 \\ \text{Im}z = 4 \end{matrix}$$

$$c) (5+2j)^2 = 25 + 20j + 4j^2 = 21 + 20j \quad \begin{matrix} \text{Re}z = 21 \\ \text{Im}z = 20 \end{matrix}$$

$$d) (3+2j)^3 = 27 + 54j + 36j^2 + 8j^3 = 27 + 54j - 36 - 8j = -9 + 46j \quad \begin{matrix} \text{Re}z = -9 \\ \text{Im}z = 46 \end{matrix}$$

$$e) \frac{1+j}{1+j\sqrt{3}} \cdot \frac{1-j\sqrt{3}}{1-j\sqrt{3}} = \frac{(1+j)(1-j\sqrt{3})}{1-3j^2} = \frac{1-j\sqrt{3}+j-j^2\sqrt{3}}{1-3j^2} = \frac{1+(1-\sqrt{3})j-\sqrt{3}}{1+3} =$$

$$= \frac{(1-\sqrt{3})}{4} + \frac{(1-\sqrt{3})}{4}j \quad \begin{matrix} \text{Re}z = \frac{1-\sqrt{3}}{4} \\ \text{Im}z = \frac{1-\sqrt{3}}{4} \end{matrix}$$

Zad. 2

a) $z^2 + 4z + 5 = 0$

$$\Delta = 16 - 20 = -4 = -1 \cdot 4 = j^2 \cdot 4 \quad \sqrt{\Delta} = 2j$$

$$z_1 = \frac{-4 - 2j}{2} = -2 - j \quad \begin{matrix} \text{Re} z = -2 \\ \text{Im} z = -1 \end{matrix}$$

$$z_2 = \frac{-4 + 2j}{2} = -2 + j \quad \begin{matrix} \text{Re} z = -2 \\ \text{Im} z = 1 \end{matrix}$$

d) $z^2 + 2(1+j)z + 2j = 0$

$$\Delta = 4(1+j)^2 - 8j = 4 + 8j + 4j^2 - 8j = 4 - 4 = 0$$

$$z_0 = \frac{-b}{2a} = \frac{-2(1+j)}{2} = -1 - j \quad \begin{matrix} \text{Re} z = -1 \\ \text{Im} z = -1 \end{matrix}$$

e) $z^2 + (1+4j)z - (5+j) = 0$

$$\Delta = (1+4j)^2 + 4(5+j) = 1 + 8j - 16 + 20 + 4j = 5 + 12j$$

$$\sqrt{\Delta} = \sqrt{5+12j}$$

$$\sqrt{5+12j} = x + jy \quad \parallel \cdot ()^2$$

$$5 + 12j = x^2 + 2xyj - y^2$$

$$\begin{cases} x^2 - y^2 = 5 \\ 2xy = 12 \end{cases}$$

$$xy = 6$$

$$y = \frac{6}{x}$$

$$x^2 - \frac{36}{x^2} = 5 \quad \parallel \cdot x^2$$

$$x^4 - 36 = 5x^2$$

$$x^4 - 5x^2 - 36 = 0$$

$$t^2 - 5t - 36 = 0 \quad t \geq 0$$

b) $z^2 + z + 1 = 0$

$$\Delta = 1 - 4 = -3 = -1 \cdot 3 = 3j^2$$

$$\sqrt{\Delta} = j\sqrt{3}$$

$$z_1 = \frac{-1 - j\sqrt{3}}{2} = -\frac{1}{2} - \frac{\sqrt{3}}{2}j$$

$$z_2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}j$$

c) $z^2 = j$

$$z = \sqrt{j}$$

$$\sqrt{j} = x + jy \quad \parallel ()^2$$

$$j = x^2 + 2xyj + j^2y^2$$

$$j = x^2 + 2xyj - y^2$$

$$R \ni \begin{cases} 0 = x^2 - y^2 \\ 1 = 2xy \end{cases}$$

$$y = \frac{1}{2x}$$

$$x^2 - \left(\frac{1}{2x}\right)^2 = 0$$

$$x^2 - \frac{1}{4x^2} = 0 \quad \parallel \cdot 4x^2$$

$$4x^4 - 1 = 0$$

$$x^4 = \frac{1}{4}$$

$$x^2 = \frac{1}{2}$$

$$\vee x^2 = -\frac{1}{2}$$

$$x = \frac{\sqrt{2}}{2} \quad \vee \quad x = -\frac{\sqrt{2}}{2}$$

sym. , bo $x \in \mathbb{R}$

$$y = \frac{\sqrt{2}}{2} \quad \vee \quad y = -\frac{\sqrt{2}}{2}$$

$$ODP: z_1 = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j, z_2 = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j$$

$$\Delta = 25 + 144 = 13^2 \quad \sqrt{\Delta} = 13$$

$$t_1 = \frac{5-13}{2} = -4 \quad t_2 = 9$$

$$\text{odreazum} \quad x_1 = 3 \vee x_2 = -3$$

$$y_1 = 2 \vee y_2 = -2$$

$$\sqrt{\Delta} = 3 + 2j \vee -\sqrt{\Delta} = -3 - 2j$$

$$Z_1 = \frac{-(1+4j) - (3+2j)}{2} = \frac{-1-4j-3-2j}{2} = \frac{-6j-4}{2} = -3j-2$$

$$Z_2 = \frac{-1-4j+3+2j}{2} = \frac{2-2j}{2} = 1-j$$

Zad. 6

$$z = -j \quad \begin{matrix} x=0 \\ y=-1 \end{matrix}$$

$$|z| = \sqrt{0^2 + (-1)^2} = 1$$

$$\begin{cases} \cos \varphi = \frac{0}{1} = 0 \\ \sin \varphi = \frac{-1}{1} = -1 \end{cases} \Rightarrow \varphi = \frac{3\pi}{2}$$

$$Z = 1(\cos \frac{3\pi}{2} + j \sin \frac{3\pi}{2})$$

$$z = -1 \quad \begin{matrix} x=-1 \\ y=0 \end{matrix}$$

$$|z| = \sqrt{0^2 + (-1)^2} = 1$$

$$\begin{cases} \cos \varphi = \frac{-1}{1} = -1 \\ \sin \varphi = \frac{0}{1} = 0 \end{cases} \Rightarrow \varphi = \pi$$

$$Z = 1(\cos \pi + j \sin \pi)$$

$$c) (-1+j)^{12}$$

$$z = -1+j \quad \begin{matrix} x=-1 \\ y=1 \end{matrix}$$

$$|z| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\begin{cases} \cos \varphi = \frac{-1}{\sqrt{2}} = -\frac{\sqrt{2}}{2} \\ \sin \varphi = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \end{cases} \Rightarrow \varphi = \frac{3\pi}{4}$$

$$\begin{aligned} (-1+j)^{12} &= (\sqrt{2})^{12} (\cos(12 \cdot \frac{3\pi}{4}) + j \sin(12 \cdot \frac{3\pi}{4})) = \\ &= 64(\cos 9\pi + j \sin 9\pi) = 64(\cos \pi + j \sin \pi) \end{aligned}$$

$$d) \frac{\sqrt{3}+j}{1+j\sqrt{3}} = \frac{|Z_1|}{|Z_2|} (\cos \varphi)$$

$$Z_1 = \sqrt{3}+j \quad \begin{matrix} x_1=\sqrt{3} \\ y_1=1 \end{matrix} \quad |Z_1|=2 \quad Z_2 = 1+j\sqrt{3} \quad \begin{matrix} x_2=1 \\ y_2=\sqrt{3} \end{matrix} \quad |Z_2|=2$$

$$\begin{cases} \cos \varphi_1 = \frac{\sqrt{3}}{2} \\ \sin \varphi_2 = \frac{1}{2} \end{cases} \Rightarrow \varphi = \frac{\pi}{6} \quad \begin{cases} \cos \varphi = \frac{1}{2} \\ \sin \varphi = \frac{\sqrt{3}}{2} \end{cases} \Rightarrow \varphi = \frac{\pi}{3}$$

Zad. 4

$$A) \frac{(1+j)^3}{(\sqrt{3}+j)^3} = \frac{(\sqrt{2})^3 \cdot (\cos(3 \cdot \frac{\pi}{4}) + j \sin(3 \cdot \frac{\pi}{4}))}{2^3 \cdot (\cos(3 \cdot \frac{\pi}{6}) + j \sin(3 \cdot \frac{\pi}{6}))} = \frac{\sqrt{2} (\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4})}{4 (\cos \frac{\pi}{2} + j \sin \frac{\pi}{2})} = \frac{\sqrt{2} (-\frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2})}{4 (0 + j)} =$$

$$Z_1 = 1+j \quad \begin{matrix} x_1=1 \\ y_1=1 \end{matrix} \quad |Z_1| = \sqrt{1^2+1^2} = \sqrt{2}$$

$$\begin{cases} \cos \varphi_1 = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \\ \sin \varphi_2 = \frac{\sqrt{2}}{2} \end{cases} \quad \varphi_2 = \frac{\pi}{4}$$

$$Z_2 = \sqrt{3}+j \quad \begin{matrix} x_1=\sqrt{3} \\ y_1=1 \end{matrix} \quad |Z_2|=2$$

$$\begin{cases} \cos \varphi_2 = \frac{\sqrt{3}}{2} \\ \sin \varphi_2 = \frac{1}{2} \end{cases} \Rightarrow \varphi_2 = \frac{\pi}{6}$$

$$= \frac{j-1}{4j} \cdot \frac{j}{j} = \frac{j^2-j}{4j} = \frac{-1-j}{-4j} = \frac{1+j}{4j}$$

$$\frac{20\pi}{3} = \frac{18\pi}{3} + \frac{2\pi}{3}$$

$$-\frac{20\pi}{4} = -5\pi$$

$$B) \left(\frac{1+j\sqrt{3}}{1-j} \right)^{20} = \frac{(1+j\sqrt{3})^{20}}{(1-j)^{20}} = \frac{2^{20} (\cos(20 \cdot \frac{\pi}{3}) + j \sin(20 \cdot \frac{\pi}{3}))}{\sqrt{2}^{20} (\cos(20 \cdot (-\frac{\pi}{4})) + j \sin(20 \cdot (-\frac{\pi}{4})))} =$$

$$Z_1 = 1+j\sqrt{3} \quad \begin{matrix} x_1=1 \\ y_1=\sqrt{3} \end{matrix} \quad |Z_1| = \sqrt{3+1} = 2$$

$$\begin{cases} \cos \varphi_1 = \frac{1}{2} \\ \sin \varphi_1 = \frac{\sqrt{3}}{2} \end{cases} \Rightarrow \varphi_1 = \frac{\pi}{3}$$

$$Z_2 = 1-j \quad \begin{matrix} x_2=1 \\ y_2=-1 \end{matrix}$$

$$|Z_2| = \sqrt{2}$$

$$\begin{cases} \cos \varphi_2 = \frac{\sqrt{2}}{2} \\ \sin \varphi_2 = -\frac{\sqrt{2}}{2} \end{cases} \Rightarrow \varphi_2 = -\frac{\pi}{4}$$

$$= \frac{2^{20} (\cos \frac{20\pi}{3} + j \sin \frac{20\pi}{3})}{2^{10} (\cos(-\pi) + j \sin(-\pi))} = 2^{10} \left(\frac{-\frac{1}{2} + j \frac{\sqrt{3}}{2}}{-1 + j \cdot 0} \right) =$$

$$= 1024 \cdot \left(\frac{1}{2} - \frac{\sqrt{3}}{2}j \right) = 512 - 512\sqrt{3}j$$

$$\text{Rez} = 512$$

$$\text{Imz} = -512\sqrt{3}$$

$$C) (\sqrt{3}-j)^{30} = 2^{30} (\cos(30 \cdot (-\frac{\pi}{6})) + j \sin(30 \cdot (-\frac{\pi}{6}))) =$$

$$= 2^{30} (\cos(-5\pi) + j \sin(-5\pi)) =$$

$$= 2^{30} (\cos(-\pi) + j \sin(-\pi)) =$$

$$= 2^{30} (0 - 1) = -2^{30}$$

$$Z = \sqrt{3} - j \quad \begin{matrix} x = \sqrt{3} \\ y = -1 \end{matrix}$$

$$|Z| = \sqrt{3+1} = 2$$

$$\begin{cases} \cos \varphi = \frac{\sqrt{3}}{2} \\ \sin \varphi = -\frac{1}{2} \end{cases} \Rightarrow \varphi = -\frac{\pi}{6}$$

$$D) \frac{(1+j)^{12} \cdot (-j)^{10}}{(\sqrt{3}+j)^2 (1-j\sqrt{3})^4} = \frac{(\sqrt{2})^{12} (\cos(12 \cdot \frac{\pi}{4}) + j \sin(12 \cdot \frac{\pi}{4})) \cdot (j^2)^5}{2^2 (\cos(2 \cdot \frac{\pi}{6}) + j \sin(2 \cdot \frac{\pi}{6})) \cdot 2^4 (\cos(4 \cdot (-\frac{\pi}{3})) + j \sin(4 \cdot (-\frac{\pi}{3})))} =$$

$$Z_1 = 1+j \quad \begin{matrix} x_1=1 \\ y_1=1 \end{matrix}$$

$$|Z_1| = \sqrt{2}$$

$$= \frac{1 \cdot (\cos 3\pi + j \sin 3\pi) \cdot (-1)^5}{1 \cdot 64 (\cos \frac{\pi}{3} + j \sin \frac{\pi}{3}) (\cos(-\frac{4\pi}{3}) + j \sin(-\frac{4\pi}{3}))} =$$

$$\begin{cases} \cos \varphi_1 = \frac{\sqrt{2}}{2} \\ \sin \varphi_1 = \frac{\sqrt{2}}{2} \end{cases} \Rightarrow \varphi_1 = \frac{\pi}{4}$$

$$= \frac{-(\cos \pi + j \sin \pi)}{(\frac{1}{2} + \frac{\sqrt{3}}{2}j) (\cos(\frac{2\pi}{3}) + j \sin \frac{2\pi}{3})} =$$

$$Z_2 = \sqrt{3} + j \quad \begin{matrix} x_2=\sqrt{3} \\ y_2=1 \end{matrix}$$

$$= \frac{-(-1 + j \cdot 0)}{\frac{1}{2}(1 + \sqrt{3}j) (-\frac{1}{2} + \frac{\sqrt{3}}{2}j)} =$$

$$|Z_2| = 2$$

$$= \frac{1}{-\frac{1}{4}(1 + \sqrt{3}j)(1 - \sqrt{3}j)} =$$

$$\begin{cases} \cos \varphi_2 = \frac{\sqrt{3}}{2} \\ \sin \varphi_2 = \frac{1}{2} \end{cases} \Rightarrow \varphi_2 = \frac{\pi}{6}$$

$$= \frac{-4}{(1 - 3j^2)} = -1$$

$$Z_3 = 1 - \sqrt{3}j \quad \begin{matrix} x_3=1 \\ y_3=-\sqrt{3} \end{matrix}$$

$$|Z_3| = 2$$

$$\begin{cases} \cos \varphi_3 = \frac{1}{2} \\ \sin \varphi_3 = -\frac{\sqrt{3}}{2} \end{cases} \Rightarrow \varphi_3 = -\frac{\pi}{3}$$

$$E) \frac{(\sqrt{3}+j)^5 \cdot (1-\sqrt{3}j)^4}{(\sqrt{3}-j)^4 \cdot (1+\sqrt{3}j)^5} = \frac{2^5 \cdot (\cos \frac{5\pi}{6} + j \sin \frac{5\pi}{6}) \cdot 2^4 \cdot (\cos(-\frac{\pi}{3}) + j \sin(-\frac{\pi}{3}))}{2^3 \cdot (\cos(-\frac{\pi}{6}) + j \sin(-\frac{\pi}{6})) \cdot 2^5 \cdot (\cos \frac{5\pi}{3} + j \sin \frac{5\pi}{3})} =$$

$$Z_1 = \sqrt{3} + j \quad \begin{matrix} x_1 = \sqrt{3} \\ y_1 = 1 \end{matrix}$$

$$|Z_1| = 2$$

$$\begin{cases} \cos \varphi_1 = \frac{\sqrt{3}}{2} \\ \sin \varphi_1 = \frac{1}{2} \end{cases} \Rightarrow \varphi_1 = \frac{\pi}{6}$$

$$Z_2 = 1 - \sqrt{3}j \quad \begin{matrix} x_2 = 1 \\ y_2 = -\sqrt{3} \end{matrix}$$

$$|Z_2| = 2$$

$$\begin{cases} \cos \varphi_2 = \frac{1}{2} \\ \sin \varphi_2 = -\frac{\sqrt{3}}{2} \end{cases} \Rightarrow \varphi_2 = -\frac{\pi}{3}$$

$$Z_3 = \sqrt{3} - j \quad \begin{matrix} x_3 = \sqrt{3} \\ y_3 = -1 \end{matrix}$$

$$|Z_3| = 2$$

$$\begin{cases} \cos \varphi_3 = \frac{\sqrt{3}}{2} \\ \sin \varphi_3 = -\frac{1}{2} \end{cases} \Rightarrow \varphi_3 = -\frac{\pi}{6}$$

$$Z_4 = 1 + \sqrt{3}j \quad \begin{matrix} x_4 = 1 \\ y_4 = \sqrt{3} \end{matrix}$$

$$|Z_4| = 2$$

$$\begin{cases} \cos \varphi_4 = \frac{1}{2} \\ \sin \varphi_4 = \frac{\sqrt{3}}{2} \end{cases} \Rightarrow \varphi_4 = \frac{\pi}{3}$$

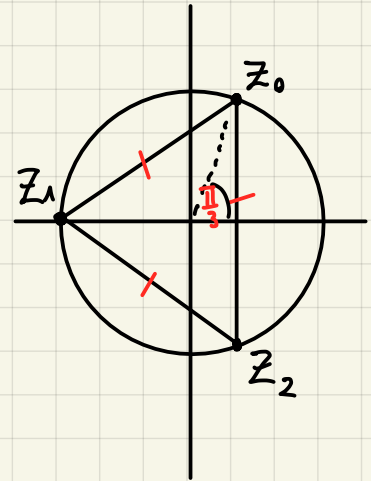
$$\begin{aligned} &= \frac{(-\frac{\sqrt{3}}{2} + \frac{1}{2}j)(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3})}{2^3(\cos \frac{5\pi}{6} + j \sin \frac{5\pi}{6})(\cos \frac{5\pi}{3} + j \sin \frac{5\pi}{3})} = \\ &= \frac{(-\frac{\sqrt{3}}{2} + \frac{1}{2}j)(-\frac{1}{2} + \frac{\sqrt{3}}{2}j)}{2^3(-\frac{\sqrt{3}}{2} + \frac{1}{2}j)(\frac{1}{2} - \frac{\sqrt{3}}{2}j)} = \\ &= \frac{1}{8} \cdot \frac{(-1 + \sqrt{3}j)}{(1 - \sqrt{3}j)} = \frac{1}{8} \cdot \frac{(-1)(1 - \sqrt{3}j)}{1 - \sqrt{3}j} = \\ &= -\frac{1}{8} \end{aligned}$$

Pieniastek $\sqrt[n]{Z}$

$$Z_k = \sqrt[n]{|Z|} \left(\cos \frac{\varphi + 2k\pi}{n} + j \sin \frac{\varphi + 2k\pi}{n} \right) \quad k=0, 1, \dots, n-1$$

$$\sqrt[3]{-1} \quad -1 = Z$$

$$|Z| = \sqrt{(-1)^2 + 0^2} = 1 \quad \begin{cases} \cos \varphi = \frac{-1}{1} = -1 \\ \sin \varphi = 0 \end{cases} \Rightarrow \varphi = \pi$$



$$Z = -1 = 1(\cos \pi + j \sin \pi)$$

$$k=0 \quad Z_0 = \sqrt[3]{1} \left(\cos \frac{\pi + 2 \cdot 0 \cdot \pi}{3} + j \sin \frac{\pi + 2 \cdot 0 \cdot \pi}{3} \right) = \cos \frac{\pi}{3} + j \sin \frac{\pi}{3}$$

$$k=1 \quad Z_1 = \cos \pi + j \sin \pi$$

$$k=2 \quad Z_2 = \cos \frac{5\pi}{3} + j \sin \frac{5\pi}{3}$$

Przykład

$$Z^5 = 1 \quad \begin{matrix} x=1 \\ y=0 \end{matrix} \quad |Z|=1 \quad \begin{cases} \cos \varphi = \frac{1}{1} = 1 \\ \sin \varphi = \frac{0}{1} = 0 \end{cases} \Rightarrow \varphi = 0$$

$$Z = 1(\cos 0 + j \sin 0)$$

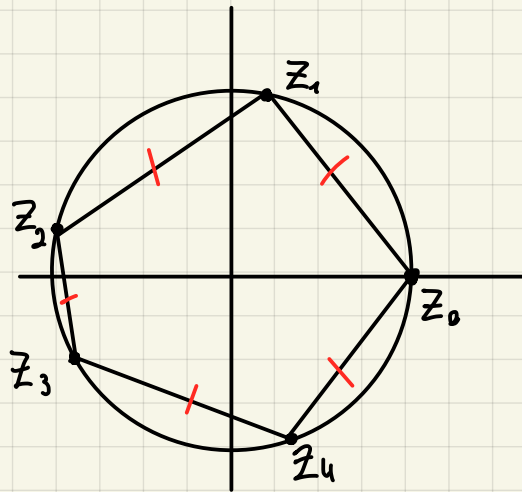
$$k=0 \quad Z_0 = \sqrt[5]{1} (\cos 0 + j \sin 0)$$

$$k=1 \quad Z_1 = \sqrt[5]{1} \left(\cos \frac{2\pi}{5} + j \sin \frac{2\pi}{5} \right)$$

$$k=2 \quad Z_2 = \sqrt[5]{1} \left(\cos \frac{4\pi}{5} + j \sin \frac{4\pi}{5} \right)$$

$$k=3 \quad Z_3 = \sqrt[5]{1} \left(\cos \frac{6\pi}{5} + j \sin \frac{6\pi}{5} \right)$$

$$k=4 \quad Z_4 = \sqrt[5]{1} \left(\cos \frac{8\pi}{5} + j \sin \frac{8\pi}{5} \right)$$



$$Z^4 = 1 - j \quad |Z| = \sqrt{2} \quad \begin{cases} \cos \varphi = \frac{\sqrt{2}}{2} \\ \sin \varphi = -\frac{\sqrt{2}}{2} \end{cases} \Rightarrow \varphi = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$

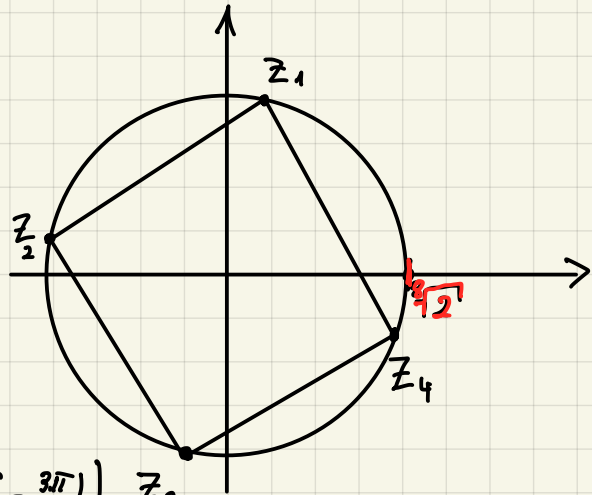
$x=1 \quad y=-1$

$$Z_0 = \sqrt[8]{2} \cdot (\cos \frac{7\pi}{16} + j \sin \frac{7\pi}{16})$$

$$Z_1 = \sqrt[8]{2} (\cos \frac{15\pi}{16} + j \sin \frac{15\pi}{16})$$

$$Z_2 = \sqrt[8]{2} (\cos \frac{23\pi}{16})$$

$$Z_3 = \sqrt[8]{2} (\cos \frac{31\pi}{16})$$



$$Z^3 = \frac{(1-j)^2}{(1+j)^3} = \frac{(\sqrt{2})^2}{(\sqrt{2})^3} (\cos(\frac{7\pi}{2} - \frac{3\pi}{4}) + j \sin(\frac{7\pi}{2} - \frac{3\pi}{4})) = Z_3$$

$$C_1 = 1 - j \quad \begin{matrix} x_1=1 \\ y_1=-1 \end{matrix}$$

$$C_2 = 1 - j \quad \begin{matrix} x_2=1 \\ y_2=-1 \end{matrix}$$

$$|C_1| = \sqrt{2}$$

$$|C_2| = \sqrt{2}$$

$$\begin{cases} \cos \varphi_1 = \frac{\sqrt{2}}{2} \\ \sin \varphi_1 = -\frac{\sqrt{2}}{2} \end{cases} \Rightarrow \varphi_1 = \frac{7\pi}{4}$$

$$\begin{cases} \cos \varphi_2 = \frac{\sqrt{2}}{2} \\ \sin \varphi_2 = -\frac{\sqrt{2}}{2} \end{cases} \Rightarrow \varphi_2 = \frac{7\pi}{4}$$

$$\begin{aligned} &\rightarrow \frac{1}{\sqrt{2}} (\cos \frac{11\pi}{4} + j \sin \frac{11\pi}{4}) = \\ &= \frac{\sqrt{2}}{2} (\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4}) = C_3 \end{aligned}$$

$$Z_0 = \sqrt[3]{\frac{\sqrt{2}}{2}} \cdot \cos(\frac{\pi}{4})$$

$$Z_1 = \sqrt[3]{\frac{\sqrt{2}}{2}} \cdot \cos(\frac{11\pi}{12})$$

$$Z_2 = \sqrt[3]{\frac{\sqrt{2}}{2}} \cdot \cos(\frac{19\pi}{12})$$

